ACTIVATED CHARCOAL

Within 1 h of ingestion

Protect airway

CHEMICAL NAME Activated carbon (finely powdered, highly porous carbon with surface area 950–2,000 m²/g)

CLASS Gastrointestinal adsorbent (poisoning)

TRADE NAMES Charcodote · Actidose-Aqua · Carbomix · Liqui-Char

THERAPEUTIC USES / INDICATIONS

- Acute oral poisoning within 1 h of ingestion of a potentially toxic amount of an adsorbable substance (e.g. paracetamol, aspirin, TCAs, antiepileptics, digoxin).
- Multiple-dose activated charcoal for life-threatening carbamazepine, phenobarbital, dapsone, quinine, theophylline poisoning.

ADMINISTRATION

- **PO** or via **NG tube** as a slurry with water; may mix with juice to improve taste.
- Only if airway is protected (alert and able to swallow, or intubated).
- Shake well; give slowly to reduce vomiting.

DOSAGES

- **Single dose:** 50 g (adult); child 1 g/kg (max 50 g).
- **Multiple dose:** 50 g every 4 h (or 25 g every 2 h) — with antiemetic, monitor bowel sounds.

PHARMACOKINETICS

- Not absorbed; excreted in faeces (black).

PHARMACODYNAMICS

- Adsorbs toxins in the GI tract → prevents systemic absorption; multiple doses enhance elimination by interrupting enterohepatic circulation ("gut dialysis").

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Black stools.
- Vomiting.
- Constipation.

ADVERSE EFFECTS (SERIOUS)

- Aspiration pneumonitis (if airway not protected or vomiting).
- Bowel obstruction (multiple doses).
- Corneal abrasion (if spilled in eyes).

CONTRAINDICATIONS & CAUTIONS

- Unprotected airway (reduced consciousness without intubation).
- Ingestion of substances not adsorbed: iron, lithium, alcohols, acids, alkalis/corrosives, petroleum products, cyanide (poor), heavy metals.
- GI obstruction, perforation, recent GI surgery.
- Need for endoscopy (corrosives).

KEY INTERACTIONS

- Adsorbs oral antidotes and drugs given at the same time (e.g. oral acetylcysteine, methionine) — separate.

NURSING MANAGEMENT

- Confirm time and substance ingested; check airway and GCS.
- Sit patient upright; give slowly; monitor for vomiting/aspiration.
- Record stools; laxative not routinely needed.
- Contact poisons information centre.

PATIENT EDUCATION

- Stools will be black for a few days.
- Explain charcoal binds poison in the stomach to stop it being absorbed.

KEY POINTS

- Adsorbs toxins; best within 1 h.
- Airway must be protected.
- 50 g adult; 1 g/kg child.
- Useless for iron, lithium, alcohols, corrosives.